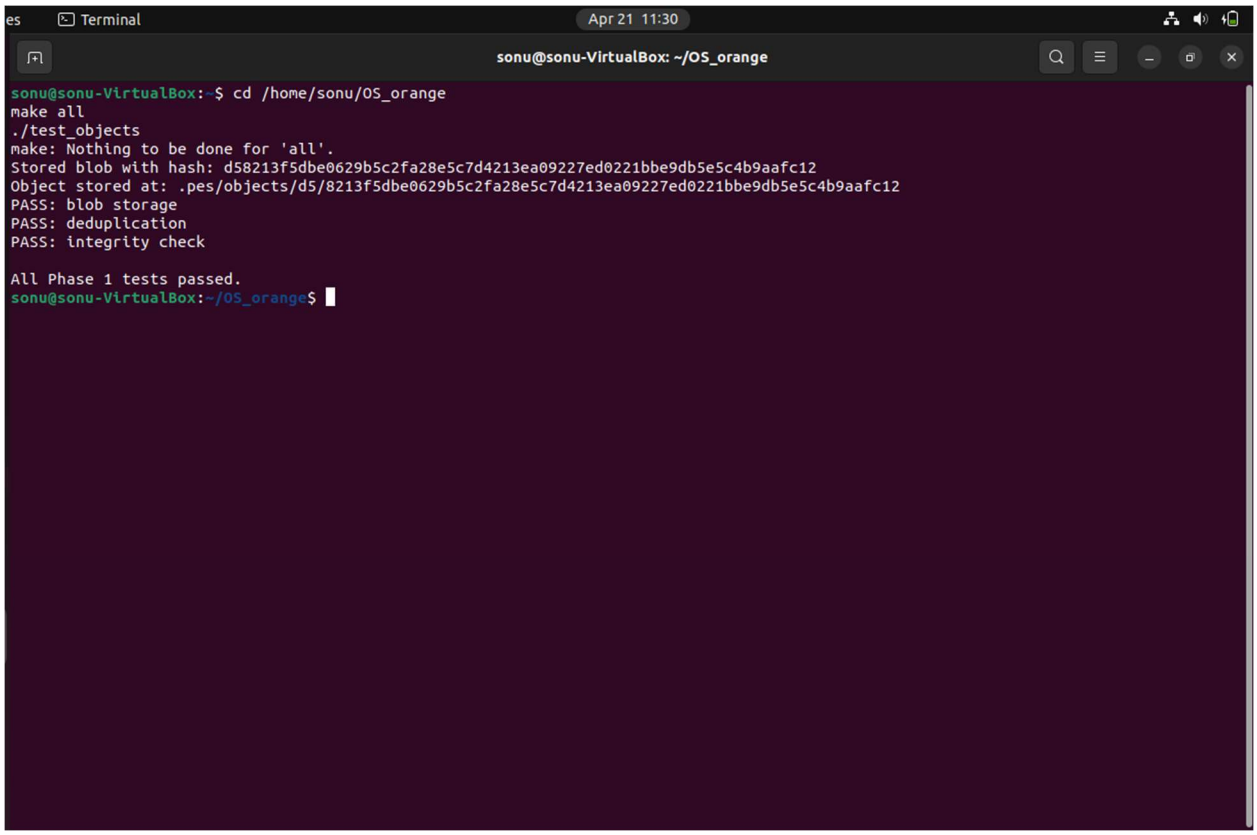
OS ORANGE 2 REPORT
Building PES-VCS — A Version Control System from Scratch

PHASE 1:

Screenshot 1A



Screenshot 1B

```
sonu@sonu-VirtualBox:~$ cd /home/sonu/OS_orange
make all
./test_objects
make: Nothing to be done for 'all'.
Stored blob with hash: d58213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
Object stored at: .pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
PASS: blob storage
PASS: deduplication
PASS: integrity check

All Phase 1 tests passed.
sonu@sonu-VirtualBox:~/OS_orange$ find .pes/objects -type f | sort
.pes/objects/25/ef1fa07ea68a52f800dc80756ee6b7ae34b337afedb9b46a1af8e11ec4f476
.pes/objects/2a/594d39232787fba8eb7287418aec99c8fc2ecdaf5aaf2e650eda471e566fcf
.pes/objects/d5/8213f5dbe0629b5c2fa28e5c7d4213ea09227ed0221bbe9db5e5c4b9aafc12
sonu@sonu-VirtualBox:~/OS_orange$
```

PHASE 2:

Screenshot 2A

```
sonu@sonu-VirtualBox:~/OS_orange$ ./test_tree
Serialized tree: 139 bytes
PASS: tree serialize/parse roundtrip
PASS: tree deterministic serialization

All Phase 2 tests passed.
```

Screenshot 2B

```
sonu@sonu-VirtualBox:~/OS_orange$ xxd .pes/objects/3b/c19934c2ccc37b9582dd42105ca4a1df8e197f51c99a3ef71628231437fdbf | head -20
00000000: 7472 6565 2039 3600 3130 3036 3434 2062  tree 96.100644 b
00000010: 7965 2e74 7874 00e6 7ed6 6bcb 4a70 8250  ye.txt..~.k.Jp.P
00000020: a89f 315d 4d8b b927 0334 3c84 df83 4438  ..1]M..'.4<...D8
00000030: 880e 13a6 8652 bc31 3030 3634 3420 6865  .....R.100644 he
00000040: 6c6c 6f2e 7478 7400 6622 4663 d23e 6f4d  llo.txt.f"Fc.>oM
00000050: 9de9 e2c7 e6d8 7643 05a9 2a38 30a1 a52d  ......vC..*80..-
00000060: 3d5f 4aa8 007b 5c39                      =_J..{\9
sonu@sonu-VirtualBox:~/OS_orange$
```

PHASE 3:

Screenshot 3A

```
es  Terminal                    Apr 21  11:59
sonu@sonu-VirtualBox: ~/OS_orange
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  file2.txt

sonu@sonu-VirtualBox:~/OS_orange$ cat .pes/index
cat: .pes/index: No such file or directory
sonu@sonu-VirtualBox:~/OS_orange$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_sequence.sh
  untracked:  object.c
  untracked:  commit.h
  untracked:  test_tree.c
  untracked:  index.h
  untracked:  pes.h
  untracked:  test_objects
  untracked:  test_tree
  untracked:  tree.c
  untracked:  tree.h
  untracked:  test_objects.c
  untracked:  Makefile
  untracked:  index.c
  untracked:  README.md
  untracked:  commit.c
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  pesdbg

sonu@sonu-VirtualBox:~/OS_orange$
```

Screenshot 3B

```
es  Terminal                    Apr 21  11:59
sonu@sonu-VirtualBox: ~/OS_orange

sonu@sonu-VirtualBox:~/OS_orange$ cat .pes/index
cat: .pes/index: No such file or directory
sonu@sonu-VirtualBox:~/OS_orange$ ./pes init
echo "hello" > file1.txt
echo "world" > file2.txt
./pes add file1.txt file2.txt
./pes status
Initialized empty PES repository in .pes/
Staged changes:
  staged:     file1.txt
  staged:     file2.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  untracked:  test_sequence.sh
  untracked:  object.c
  untracked:  commit.h
  untracked:  test_tree.c
  untracked:  index.h
  untracked:  pes.h
  untracked:  test_objects
  untracked:  test_tree
  untracked:  tree.c
  untracked:  tree.h
  untracked:  test_objects.c
  untracked:  Makefile
  untracked:  index.c
  untracked:  README.md
  untracked:  commit.c
  untracked:  pes.c
  untracked:  .gitignore
  untracked:  pesdbg

sonu@sonu-VirtualBox:~/OS_orange$ cat .pes/index
100644 2cf8d83d9ee29543b34a87727421fdecb7e3f3a183d337639025de576db9ebb4 1776752956 6 file1.txt
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776752956 6 file2.txt
sonu@sonu-VirtualBox:~/OS_orange$
```

PHASE 4:

Screenshot 4A

```
100644 e00c50e16a2df38f8d6bf809e181ad0248da6e6719f35f9f7e65d6f606199f7f 1776752956 6 file2.txt
sonu@sonu-VirtualBox:~/OS_orange$ ./pes init
echo "Hello" > hello.txt
./pes add hello.txt
./pes commit -m "Initial commit"

echo "World" >> hello.txt
./pes add hello.txt
./pes commit -m "Add world"

echo "Goodbye" > bye.txt
./pes add bye.txt
./pes commit -m "Add farewell"

./pes log
Initialized empty PES repository in .pes/
Committed: beb24c2c1c98... Initial commit
Committed: c6c0312a7f49... Add world
Committed: 5908ec7e7f5f... Add farewell
commit 5908ec7e7f5fa8464581232fd08ff13b44165a9ea4b7d75fab9e8955528df2d4
Author: PES User <pes@localhost>
Date:   1776753382

    Add farewell

commit c6c0312a7f498fc554ef757c9374db3c7bea7c18063b6d7aa269102f78c7c565
Author: PES User <pes@localhost>
Date:   1776753382

    Add world

commit beb24c2c1c985dc784502bf0d027ab34e2a1526bed36598ad25f9a862070135b
Author: PES User <pes@localhost>
Date:   1776753381

    Initial commit

commit 2b797df2e2ed762fb760b54e6044a13cb01b8a8a31b522ab10b203c545a8748f
Author: PES User <pes@localhost>
Date:   1776753334
```

es Terminal Apr 21 12:06

sonu@sonu-VirtualBox: ~/OS_orange

```
Committed: beb24c2c1c98... Initial commit
Committed: c6c0312a7f49... Add world
Committed: 5908ec7e7f5f... Add farewell
commit 5908ec7e7f5fa8464581232fd08ff13b44165a9ea4b7d75fab9e8955528df2d4
Author: PES User <pes@localhost>
Date:   1776753382

    Add farewell

commit c6c0312a7f498fc554ef757c9374db3c7bea7c18063b6d7aa269102f78c7c565
Author: PES User <pes@localhost>
Date:   1776753382

    Add world

commit beb24c2c1c985dc784502bf0d027ab34e2a1526bed36598ad25f9a862070135b
Author: PES User <pes@localhost>
Date:   1776753381

    Initial commit

commit 2b797df2e2ed762fb760b54e6044a13cb01b8a8a31b522ab10b203c545a8748f
Author: PES User <pes@localhost>
Date:   1776753334

    Add farewell

commit 07085fdff0cc43d376320e161935fca0ea5b3421c3859b34f9da1ee2aaa90c91
Author: PES User <pes@localhost>
Date:   1776753334

    Add world

commit 7e6260c0eaa252c162d602a2aebbd8ad300a38bc4d676446a3d06ee1adc93b76
Author: PES User <pes@localhost>
Date:   1776753334

    Initial commit

sonu@sonu-VirtualBox:~/OS_orange$
```

Screenshot 4B

```
Date:   1776753381

    Initial commit

commit 2b797df2e2ed762fb760b54e6044a13cb01b8a8a31b522ab10b203c545a8748f
Author: PES User <pes@localhost>
Date:   1776753334

    Add farewell

commit 07085fdff0cc43d376320e161935fca0ea5b3421c3859b34f9da1ee2aaa90c91
Author: PES User <pes@localhost>
Date:   1776753334

    Add world

commit 7e6260c0eaa252c162d602a2aebbd8ad300a38bc4d676446a3d06ee1adc93b76
Author: PES User <pes@localhost>
Date:   1776753334

    Initial commit

sonu@sonu-VirtualBox:~/OS_orange$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/07/085fdff0cc43d376320e161935fca0ea5b3421c3859b34f9da1ee2aaa90c91
.pes/objects/2b/797df2e2ed762fb760b54e6044a13cb01b8a8a31b522ab10b203c545a8748f
.pes/objects/3b/c19934c2ccc37b9582dd42105ca4a1df8e197f51c99a3ef71628231437fdbf
.pes/objects/59/08ec7e7f5fa8464581232fd08ff13b44165a9ea4b7d75fab9e8955528df2d4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/7e/6260c0eaa252c162d602a2aebbd8ad300a38bc4d676446a3d06ee1adc93b76
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/be/b24c2c1c985dc784502bf0d027ab34e2a1526bed36598ad25f9a862070135b
.pes/objects/c6/c0312a7f498fc554ef757c9374db3c7bea7c18063b6d7aa269102f78c7c565
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
sonu@sonu-VirtualBox:~/OS_orange$
```

Screenshot 4C

```
es   Terminal                    Apr 21  12:07
sonu@sonu-VirtualBox: ~/OS_orange
commit 2b797df2e2ed762fb760b54e6044a13cb01b8a8a31b522ab10b203c545a8748f
Author: PES User <pes@localhost>
Date:   1776753334

    Add farewell

commit 07085fdff0cc43d376320e161935fca0ea5b3421c3859b34f9da1ee2aaa90c91
Author: PES User <pes@localhost>
Date:   1776753334

    Add world

commit 7e6260c0eaa252c162d602a2aebbd8ad300a38bc4d676446a3d06ee1adc93b76
Author: PES User <pes@localhost>
Date:   1776753334

    Initial commit

sonu@sonu-VirtualBox:~/OS_orange$ find .pes -type f | sort
.pes/HEAD
.pes/index
.pes/objects/05/c8d777d0be85748b6788d1b5caf7d69754e6e68060c265ca21cbd6c05e5f43
.pes/objects/07/085fdff0cc43d376320e161935fca0ea5b3421c3859b34f9da1ee2aaa90c91
.pes/objects/2b/797df2e2ed762fb760b54e6044a13cb01b8a8a31b522ab10b203c545a8748f
.pes/objects/3b/c19934c2ccc37b9582dd42105ca4a1df8e197f51c99a3ef7162823143 7fdbf
.pes/objects/59/08ec7e7f5fa8464581232fd08ff13b44165a9ea4b7d75fab9e8955528df2d4
.pes/objects/66/224663d23e6f4d9de9e2c7e6d8764305a92a3830a1a52d3d5f4aa8007b5c39
.pes/objects/7e/6260c0eaa252c162d602a2aebbd8ad300a38bc4d676446a3d06ee1adc93b76
.pes/objects/a6/882ee4afdc855ea4e39f4e0fc77d90e73a6f72deb383e11934310a10a030f6
.pes/objects/bd/9b769bb4d9910c31b4f0e99e4375fffea35c2055130c248ff131f78c626442
.pes/objects/be/b24c2c1c985dc784502bf0d027ab34e2a1526bed36598ad25f9a862070135b
.pes/objects/c6/c0312a7f498fc554ef757c9374db3c7bea7c18063b6d7aa269102f78c7c565
.pes/objects/d8/f28203071e10a3bde605928368bc5b51871287a867d8d10382967099dde814
.pes/objects/e6/7ed66bcb4a708250a89f315d4d8bb92703343c84df834438880e13a68652bc
.pes/refs/heads/main
sonu@sonu-VirtualBox:~/OS_orange$ cat .pes/refs/heads/main
cat .pes/HEAD
5908ec7e7f5fa8464581232fd08ff13b44165a9ea4b7d75fab9e8955528df2d4
ref: refs/heads/main
sonu@sonu-VirtualBox:~/OS_orange$
```

FINAL:

```
es  Terminal                                   Apr 21 12:14
sonu@sonu-VirtualBox: ~/OS_orange
00000060: 3d5f 4aa8 007b 5c39                      =_J..{\9
sonu@sonu-VirtualBox:~/OS_orange$ make test-integration
=== Running integration tests ===
bash test_sequence.sh
=== PES-VCS Integration Test ===

--- Repository Initialization ---
Initialized empty PES repository in .pes/
PASS: .pes/objects exists
PASS: .pes/refs/heads exists
PASS: .pes/HEAD exists

--- Staging Files ---
Status after add:
Staged changes:
  staged:     file.txt
  staged:     hello.txt

Unstaged changes:
  (nothing to show)

Untracked files:
  (nothing to show)


--- First Commit ---
Committed: 8a075f98fdaf... Initial commit

Log after first commit:
commit 8a075f98fdaf6ce85395bb4f3e9b7072b9d853ff7975cae25bfb7c7757b3d821
Author: PES User <pes@localhost>
Date:   1776753868

    Initial commit


--- Second Commit ---
Committed: e7c2ef53c411... Update file.txt

--- Third Commit ---
```

```
es  Terminal                                   Apr 21 12:15
sonu@sonu-VirtualBox: ~/OS_orange

    Initial commit


--- Second Commit ---
Committed: e7c2ef53c411... Update file.txt

--- Third Commit ---
Committed: fe449228f5fc... Add farewell

--- Full History ---
commit fe449228f5fcdabdaffe3b9c261f41e1bb15d40ed7c9bf5bf4b1b6d2e47f30b2
Author: PES User <pes@localhost>
Date:   1776753868

    Add farewell

commit e7c2ef53c411b1d91e7c4c063dba19b32cc8399d5c3ca0054cb1847ec1db2a4b
Author: PES User <pes@localhost>
Date:   1776753868

    Update file.txt

commit 8a075f98fdaf6ce85395bb4f3e9b7072b9d853ff7975cae25bfb7c7757b3d821
Author: PES User <pes@localhost>
Date:   1776753868

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
fe449228f5fcdabdaffe3b9c261f41e1bb15d40ed7c9bf5bf4b1b6d2e47f30b2

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
```

```
sonu@sonu-VirtualBox: ~/OS_orange
Author: PES User <pes@localhost>
Date:   1776753868

    Add farewell

commit e7c2ef53c411b1d91e7c4c063dba19b32cc8399d5c3ca0054cb1847ec1db2a4b
Author: PES User <pes@localhost>
Date:   1776753868

    Update file.txt

commit 8a075f98fdaf6ce85395bb4f3e9b7072b9d853ff7975cae25bfb7c7757b3d821
Author: PES User <pes@localhost>
Date:   1776753868

    Initial commit


--- Reference Chain ---
HEAD:
ref: refs/heads/main
refs/heads/main:
fe449228f5fcdabdaffe3b9c261f41e1bb15d40ed7c9bf5bf4b1b6d2e47f30b2

--- Object Store ---
Objects created:
10
.pes/objects/0b/d69098bd9b9cc5934a610ab65da429b525361147faa7b5b922919e9a23143d
.pes/objects/10/1f28a12274233d119fd3c8d7c7216054ddb5605f3bae21c6fb6ee3c4c7cbfa
.pes/objects/58/a67ed1c161a4e89a110968310fe31e39920ef68d4c7c7e0d6695797533f50d
.pes/objects/8a/075f98fdaf6ce85395bb4f3e9b7072b9d853ff7975cae25bfb7c7757b3d821
.pes/objects/ab/5824a9ec1ef505b5480b3e37cd50d9c80be55012cabe0ca572dbf959788299
.pes/objects/b1/7b838c5951aa88c09635c5895ef7e08f7fa1974d901ce282f30e08de0ccd92
.pes/objects/d0/7733c25d2d137b7574be8c5542b562bf48bafeaa3829f61f75b8d10d5350f9
.pes/objects/db/07a1451ca9544dbf66d769b505377d765efae7adc6b97b75cc9d2b3b3da6ff
.pes/objects/e7/c2ef53c411b1d91e7c4c063dba19b32cc8399d5c3ca0054cb1847ec1db2a4b
.pes/objects/fe/449228f5fcdabdaffe3b9c261f41e1bb15d40ed7c9bf5bf4b1b6d2e47f30b2

=== All integration tests completed ===
sonu@sonu-VirtualBox:~/OS_orange$
```

PHASE 5:

Q5.1 — Implementing pes checkout <branch>
To switch branches, three things must change in .pes/: HEAD must be updated to ref: refs/heads/<branch>, and the working directory files must be updated to match the target branch's tree. The process is: read the target branch's commit hash from .pes/refs/heads/<branch>, walk the commit's tree object recursively, then for each file in the tree write the blob contents to the working directory, removing files that don't exist in the target tree. The complexity comes from handling files that exist in the current branch but not the target — those must be deleted. You also need to update the index to match the new tree so status works correctly after checkout.

Q5.2 — Detecting dirty working directory conflicts
For each file that differs between the two branches' trees, check two things: first, compare the file's current blob hash (by reading and hashing it) against the index entry's stored hash — if they differ, the file has unstaged changes. Second, compare the index entry's hash against the current branch's tree hash for that file — if they differ, the file has staged but uncommitted changes. If

either condition is true for any file that would be overwritten by checkout, abort with an error. This requires no extra data structures — just the index and the object store.

Q5.3 — Detached HEAD and recovery
In detached HEAD state, HEAD contains a raw commit hash instead of ref: refs/heads/main. New commits are created normally and the HEAD hash is updated, but no branch pointer moves. If you switch branches or close the repo, those commits become unreachable — no branch points to them. Recovery: if you remember the hash (it was printed on commit), run git branch recover-branch <hash> to create a branch pointing to it. If you don't remember it, Git's reflog tracks recent HEAD values, so git reflog shows the lost commits. PES-VCS has no reflog, so recovery would require scanning all objects in .pes/objects/ to find commit objects and reconstruct the chain.

PHASE 6:

Q6.1 — Garbage collection algorithm
Start from all branch refs in .pes/refs/heads/. For each ref, walk the commit chain following parent pointers. For each commit, add its tree hash to a reachable set, then recursively walk the tree adding all blob and subtree hashes. Any object file in .pes/objects/ whose hash is not in the reachable set is unreachable and can be deleted. Use a HashSet for the reachable set — O(1) lookup. For 100,000 commits with 50 branches, assuming an average of 20 objects per commit (blobs + trees), you'd visit roughly 2,000,000 objects in the worst case, but with deduplication the actual unique objects visited would be far less since unchanged files share blobs across commits.

Q6.2 — GC race condition
The race: a commit operation calls tree_from_index() which writes new blob objects to the object store, but hasn't yet called head_update() to publish the commit. At this exact moment, GC runs, scans all reachable objects starting from branch refs, doesn't find the new blobs (no commit points to them yet), and deletes them. When the commit operation then writes the commit object and updates HEAD, it points to a tree that references now-deleted blobs — the repository is corrupted. Git avoids this by using a grace period: objects newer than 2 weeks are never deleted regardless of reachability, giving any in-progress operation time to complete. It also writes a FETCH_HEAD lock and checks for in-progress operations before running GC.